

Customer Journey Intelligence (JourneyIQ)

Comprehensive Project Presentation Breakdown, Predictive Machine Learning Architecture, Empirical Research Results (RQ1–RQ4), and Full Platform Review Guide.

Platform: JourneyIQ Enterprise Suite **Dataset:** RetailRocket (2.75M Events)

Tech Stack: React 19 • Vite • Next.js 16 • PyTorch • XGBoost • PostgreSQL

Architecture: Multi-Tenant SaaS • Offline Simulation Protocol

1. Executive Summary & Core Value Proposition

In modern e-commerce, businesses capture millions of digital touchpoints (clicks, product views, category browsing, cart additions, search queries). However, **over 85% of visitors remain in browsing/inactive stages**, and raw transaction rates often fall below **0.1%**. Traditional marketing automation relies on blunt, batch campaigns (broadcasting generic discount codes or blast emails) that:

- **Trigger Marketing Fatigue & Unsubscribes:** Annoying casual browsers with unsolicited emails.
- **Cannibalize Margin:** Giving unnecessary discounts to high-intent shoppers who were already going to buy.
- **Lack Timing Relevance:** Failing to understand where a customer is in their dynamic journey timeline.

JourneyIQ solves this by acting as an AI Decision Brain: It continuously models event sequences, quantifies purchase propensity & churn risk, segments shoppers into dynamic behavioral cohorts, and prescribes the **Next Best Marketing Action (NBMA)**—including explicit, non-intrusive **WAIT & SUPPRESS** decisions that preserve brand equity and marketing spend.

CLEAN BEHAVIORAL
EVENTS

2.75M

Processed RetailRocket logs

UNIQUE CUSTOMER
PROFILES

1.40M

Sessionized user journeys

RESEARCH COHORT
SPLIT

35,000

Chronological 80/20 τ cutoff

FATIGUE SUPPRESSIONS

4,391

62.7% non-intrusive wait
actions

2. End-to-End System Architecture

The platform is organized into five decoupled layers operating across Python ML pipelines, a Next.js API server, and a React Vite frontend:

Layer 1 & 2: Data Ingestion & Journey Processing

- **Ingestion:** Cleans raw `events.csv`, category hierarchy (1,669 nodes), and item stock status (36.8k catalog properties).
- **Sessionization:** Groups sequential interactions using an industry-standard **30-minute inactivity boundary**.
- **Journey State Engine:** Maps dynamic customer progress across 7 stages: *Inactive* → *Initial Engagement* → *Exploration* → *Consideration* → *High Intent* → *Conversion* → *Retention*.

Layer 3 & 4: Predictive AI & NBMA Decision Engine

- **Sequence Modeling:** GRU Deep Neural Network predicting the exact next touchpoint (View, Cart, Purchase).
- **Propensity Engine:** Journey-aware XGBoost & Logistic Regression evaluated on PR-AUC.
- **NBMA Decision Matrix:** Multi-criteria policy engine evaluating purchase probability, journey stage, churn risk, and catalog stock to select actions (Email, Discount, Cart Reminder, Wait, Suppress).

3. Research Methodology & Experimental Protocol (Paper Sec. IX-A)

No Future Peeking (Strict Chronological Cutoff): In e-commerce forecasting, random train/test splits cause severe *data leakage* (predicting past events using future observations). We enforce a strict chronological cutoff at $\tau = 80$ th percentile of dataset timestamps.

1. Training Set (70%)

24,500 visitors with all behavioral features extracted exclusively before τ .

2. Validation Set (10%)

3,500 visitors used for tuning XGBoost `scale_pos_weight = 2.0` on validation PR-AUC.

3. Held-Out Test Set (20%)

7,000 visitors evaluated on post- τ ground truth (0.07% true conversion rate).

4. Slide-by-Slide PPT Presentation Analysis

Detailed analysis of each slide from the project review presentation ([Customer-Journey-Intelligence.pptx](#)):

SLIDE 1 & 2

Title & The Core Industry Problem

Key Concept: "Clicks become useful when they lead to action." Modern stores have too many clicks but zero clarity on who to nurture, who to incentivize, and who to leave alone.

Speaker Focus: Explain that raw tracking is easy, but actionable, explainable decisioning without customer fatigue is the real challenge.

SLIDE 3

End-to-End System Architecture

Key Concept: Four-color architectural pipeline: *Green (Data Pipeline) → Purple (Intelligence & ML Models) → Orange (REST API & Web App) → Grey (Role-based users: Admin, Analyst, Manager).*

Speaker Focus: Highlight that the backend and frontend are production-ready decoupled systems with full multi-tenancy.

SLIDE 4 & 6

The Data Shape & Funnel Reality

Key Concept: 2,756,101 raw events cleaned to 2,755,641 events (removing 460 duplicates). The funnel demonstrates extreme e-commerce drop-off:

Funnel Stage	Visitor Count	Percentage of 35k Cohort	Drop-Off Rate
All Cohort Visitors	35,000	100.00%	—
≥1 Product View	34,913	99.75%	0.25% drop
≥1 Add-to-Cart	928	2.65%	97.34% drop from View
≥1 Purchase (Pre-Cutoff)	283	0.81%	69.5% drop from Cart
Post-Cutoff Future Buyers	26	0.07%	Severe Imbalance Ground Truth

SLIDE 5

The Comprehensive Model Suite

Key Concept: Covers 4 interconnected ML dimensions: Next-Event Forecasting, Behavioral Grouping, Purchase Propensity with PR-AUC, and the NBMA Decision & SHAP Attribution Layer.

SLIDE 7 & 8

The Four Core Research Questions (RQ1-RQ4)

Key Concept: Formal evaluation results on the held-out test set (\$N=7,000\$). See detailed tables in Section 5 below.

SLIDE 9 & 10 How to Read the Results & Viva Takeaways

Key Concept: *"Accuracy is not the answer when the positive class is rare."* In severe 0.07% imbalance, a dummy model predicting zero purchases achieves 99.93% accuracy but is completely useless. Trust **Macro-F1 for sequences**, **PR-AUC for propensity**, and **Fatigue Suppression count for strategy simulation**.

5. Comprehensive Machine Learning Evaluation (RQ1 to RQ4)

5.1 RQ1: Next-Event Prediction (Sequence Modeling)

Goal: Forecast whether the customer's next interaction will be a `PRODUCT_VIEW`, `ADD_TO_CART`, or `PURCHASE`.

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	Top-2 Accuracy	Evaluation Notes
1st-Order Markov Chain	0.9869	0.3298	0.3325	0.3311	1.0000	Transition probability matrix baseline
Static Logistic Regression	0.9843	0.3298	0.3316	0.3307	0.9843	Aggregated count features
Random Forest Classifier	0.9843	0.3298	0.3316	0.3307	0.9843	100 decision trees
XGBoost Classifier	0.9895	0.3298	0.3333	0.3316	0.9895	Gradient boosted trees
GRU Neural Network ★	0.9921	0.6640	0.4167	0.4653	1.0000	CHAMPION: Recurrent sequence state memory

Why GRU Won RQ1: Non-recurrent models collapse into predicting `PRODUCT_VIEW` (98%+ of events). The Gated Recurrent Unit (GRU) maintains hidden state memory across event timestamps, correctly detecting the transition signature when browsing velocity intensifies into carting and purchase.

5.2 RQ2: Customer Segmentation & Behavioral Profiling

Goal: Discover meaningful, cohesive customer clusters based on Recency, Frequency, Monetary (RFM), and Browse Velocity.

K-Means Clusters	Silhouette Score	Finding	Discovered Segment	Customer Count	% Share
K = 3	0.5531	Under-segmented	At-Risk / Inactive	23,894	68.27%
K = 4 ★	0.5606	Optimal Cluster Separation	Recent Browsers	10,058	28.74%
K = 5	0.5537	High cohesion	Cart Abandoners	672	1.92%
K = 6	0.5215	Cluster fragmentation	Champions & High Value	283	0.81%
K = 7 / 8	0.5099 / 0.5120	Over-segmentation	High Intent Cohort	93	0.27%

5.3 RQ3: Purchase Propensity Modeling (Static vs. Journey-Aware Features)

Goal: Predict post- τ conversion under extreme class imbalance (0.07% true positive rate).

Feature Set	Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC ★
Static Features	Logistic Regression ★	0.8164	0.0055	0.3889	0.0108	0.5677	0.1483 (Test Champion)
Static Features	Random Forest	0.9973	0.0000	0.0000	0.0000	0.6331	0.0148
Static Features	XGBoost	0.9974	0.0000	0.0000	0.0000	0.7473	0.0528
Journey-Aware	Logistic Regression	0.5274	0.0030	0.5556	0.0060	0.6274	0.1050
Journey-Aware	Random Forest	0.9961	0.0000	0.0000	0.0000	0.7408	0.0407
Journey-Aware	XGBoost (Validation Tuned)	0.9961	0.0000	0.0000	0.0000	0.9946 (Val AUC)	0.0322

Top Global SHAP Drivers

Features driving journey-aware predictions:

- **stage_code** : 2.7318 mean |SHAP| (Dominant signal)
- **total_events** : 0.1280 mean |SHAP|
- **recency_days** : 0.0278 mean |SHAP|
- **gru_latent_embed** : 0.0051 mean |SHAP|

Understanding the Metrics (PR-AUC vs ROC-AUC)

ROC-AUC can be deceptively high (~0.99) with 99.9% true negatives. **PR-AUC (Precision-Recall AUC)** focuses strictly on positive convertors. Static Logistic Regression achieved the highest held-out test PR-AUC (0.1483).

5.4 RQ4: Strategy Simulation & Policy Evaluation (Eq. 15 Protocol)

Goal: Counterfactual offline comparison of marketing intervention policies on 7,000 held-out visitors.

Marketing Strategy Policy	Target Precision (Prectarget)	Audience Coverage	Unnecessary Interventions	Wait & Suppress Count ★	Action Distribution Breakdown
Generic Marketing (Baseline 1)	0.0026	100.0% (1.0000)	71	0 (0.0%)	7,000 Broadcast Emails
Segment-Based Rules (Baseline 2)	0.0026	100.0% (1.0000)	285 (High Spam)	0 (0.0%)	6,701 Emails, 72 Discounts, 156 Cart Reminders, 71 Cross-Sells
Journey-Intelligence (Our System)	0.0023	33.3% (0.3333)	149 (-47.7% vs Segment)	4,391 (62.7% Spared)	4,348 WAIT, 2,397 Re-Engagement, 156 Cart Reminders, 43 Suppress

Why Generic "Wins" on Raw Prectarget in Offline Simulation: Because conversion is rare (0.07%), blasting 100% of visitors guarantees touching all convertors. However, it creates severe customer fatigue. **Journey-Intelligence intentionally issues 4,391 Wait/Suppress decisions**, preventing spam, eliminating coupon waste, and improving long-term retention.

6. Complete Platform UI & Features Walkthrough

1. Overview & Customer 360

- **Executive KPI Bar:** Conversion rate, active journeys, cart abandonment rate.
- **Customer Timeline:** Chronological clickstream displaying item views, cart additions, and duration.
- **Customer 360 Drawer:** Shows live RFM segment, churn risk, and active NBMA prescription.

2. Journey Explorer & Funnel Analytics

- **Journey Stage Distribution:** Live tracking across 7 stages.
- **Drop-Off Funnel:** Pinpointing drop-offs between View → Cart (97.3%) and Cart → Purchase (69.5%).
- **Path Analysis:** Most common multi-touch sequences.

3. Next Best Action (NBMA) Console

- **Policy Rules:** Instant discount vs cart reminder vs suppression.
- **Catalog Stock Validation:** Cross-checks item inventory before recommending products.
- **Suppression Engine:** Blocks marketing for fatigued or dormant users.

4. Experiments & Model Evaluation

- **Strategy Comparison:** Table XIII simulation & interactive ROI/Fatigue sandbox.
- **Model Evaluation:** ROC-AUC, PR-AUC, Confusion Matrix, and SHAP attribution.
- **Live Inference Simulator:** Test real-time scoring with custom user parameters.

7. Examiner Viva & Review Questions Cheat Sheet

Q1: How did you prevent data leakage during model training?

Answer: We implemented a strict chronological split using a cutoff timestamp τ (80th percentile). All customer behavioral features were calculated strictly from events occurring at or before τ , while target labels (purchase vs non-purchase) were derived exclusively from events occurring after τ . We avoided random splits which leak future interactions into the training set.

Q2: Why did you use PR-AUC instead of standard Accuracy or ROC-AUC for RQ3?

Answer: Our dataset exhibits severe class imbalance (0.07% post- τ purchase rate). In such distributions, a baseline model that predicts "no purchase" for 100% of users achieves 99.93% accuracy but has zero utility. ROC-AUC is distorted by large true-negative counts. PR-AUC directly evaluates the precision-recall trade-off on rare positive buyers.

Q3: Why did the Generic strategy win on Pretarget in RQ4, and why is Journey Intelligence still superior?

Answer: In offline holdout simulation, generic broadcast touches 100% of visitors, catching all rare buyers by brute force. However, it sends 7,000 spam messages. Journey Intelligence intentionally trades brute-force coverage for 4,391 non-intrusive WAIT/SUPPRESS decisions (62.7%), reducing unnecessary marketing interruptions by 47.7% and protecting customer lifetime value.

Q4: What is the role of SHAP in your Next Best Action decision engine?

Answer: SHAP (SHapley Additive exPlanations) provides transparent feature attribution. For each prediction, SHAP calculates the exact marginal contribution of signals like `stage_code`, `total_events`, and `recency_days`, allowing marketing analysts to understand *why* an action was recommended rather than trusting a black box.